

GhostEncoder: Stealthy backdoor attacks with dynamic triggers to pre-trained encoders in self-supervised learning

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ABSTRACT

Within the realm of computer vision, self-supervised learning (SSL) pertains to training pre-trained image encoders utilizing a substantial quantity of unlabeled images. Pre-trained image encoders can serve as feature extractors, facilitating the construction of downstream classifiers for various tasks. However, the use of SSL has led to an increase in security research related to various backdoor attacks. Currently, the trigger patterns used in backdoor attacks on SSL are mostly visible or static (sample-agnostic), making backdoors less covert and significantly affecting the attack performance. In this work, we propose GhostEncoder, the first dynamic invisible backdoor attack on SSL. Unlike existing backdoor attacks on SSL, which use visible or static trigger patterns, GhostEncoder utilizes image steganography techniques to encode hidden information into benign images and generate backdoor samples. We then fine-tune the pre-trained image encoder on a manipulation dataset to inject the backdoor, enabling downstream classifiers built upon the backdoored encoder to inherit the backdoor behavior for target downstream tasks. We evaluate GhostEncoder on three downstream tasks and results demonstrate that GhostEncoder provides practical stealthiness on images and deceives the victim model with a high attack success rate without compromising its utility. Furthermore, GhostEncoder withstands state-of-the-art defenses, including STRIP, STRIP-CL, and SSL-Cleanse.

1. Introduction

Self-supervised learning (SSL) is an emerging paradigm in machine learning that not only reduces reliance on labeled data but also enhances adversarial robustness by introducing challenges to manipulating model predictions Grill et al. (2020); He et al. (2020a); Hjelm et al. (2018); Chen et al. (2020a,c). However, SSL's performance heavily depends on abundant unlabeled data, which increases the associated computational costs. Regular users tend to adopt third-party pre-trained encoders available online to mitigate this issue. Yet, the opacity of the training process introduces new security threats. Backdoor attacks Gu et al. (2017) represent a menace within Deep Neural Network (DNNs) training, involving the malicious manipulation of prediction behavior through the contamination of a subset of training samples. Attackers implant hidden backdoors by embedding selected patterns (referred to

as backdoor triggers) into images, creating a backdoor training set, and training the model with this set Liu et al. (2018); Yao et al. (2019); Carlini and Terzis (2021). The backdoored model will covertly alter predictions to the target label when the trigger is embedded in inputs while maintaining normal behavior on benign samples.

Pre-trained encoders are employed as feature extractors for downstream classifiers in SSL. However, obtaining completely trustworthy encoders is an exceedingly challenging issue. As depicted in Fig. 1, one form of attacker involves an untrusted service provider who injects backdoors into their pre-trained encoders, which are subsequently provided to users for fine-tuning in downstream tasks. Consequently, the backdoor becomes active in the attacker's targeted downstream tasks, while the performance on non-targeted downstream tasks remains unaffected. Malicious third parties constitute another type of attacker. They download untainted image encoders, inject backdoors into them, and

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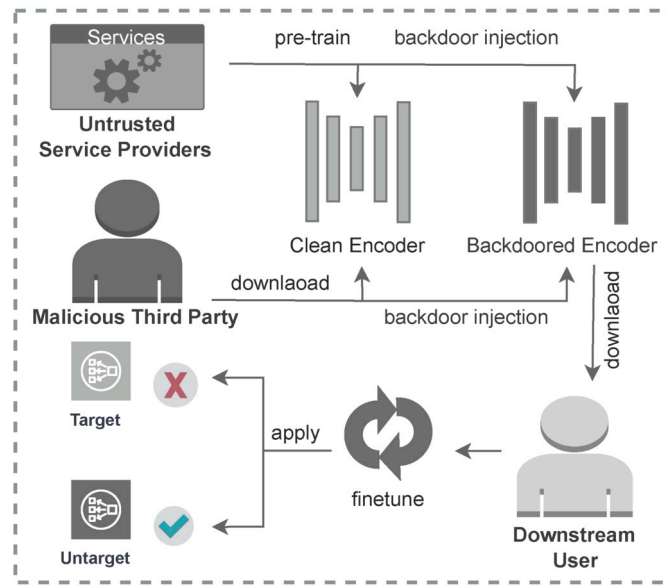


Fig. 1. The overview framework of the proposed method.

then release the compromised encoders on open-source platforms such as GitHub, Hugging Face, and Model Zoo, which makes these encoders available for unsuspecting victims to download. In particular, Liu et al. Jia et al. (2022) introduced a method that involves modifying model weights to inject backdoors into pre-trained encoders, associating trigger patterns with reference inputs from the representation space of the target class. Furthermore, there exist poison-based attack methods, assuming adversaries can contaminate a small portion of training data without exerting any control over the training process. Saha et al. Saha et al. (2022) proposed poison-based backdoor attacks on SSL. Novel approaches have also adopted this poison-based attack strategy Li et al. (2023), albeit with alterations limited to the trigger.

However, the concealment effectiveness of the trigger patterns used in these existing backdoor attacks is limited, as their backdoor triggers are sample-agnostic. This means that different backdoored samples contain the same trigger, regardless of which trigger pattern is used. As the trigger is sample-agnostic, defenders can easily reconstruct or detect the backdoor triggers based on shared behaviors among different backdoored samples, making the backdoor easily mitigated or eliminated by current defenses. For instance, in supervised scenarios, STRIP Gao et al. (2019) is effective in defending against patch-based backdoor attacks with fixed triggers. However, we observe its limited efficacy in self-supervised scenarios. Then we reveal the reasons behind the failure of STRIP in the self-supervised setting and propose a new defense called STRIP-CL. The experiments demonstrate that STRIP-CL can significantly mitigate patch-type trigger-based backdoor attacks.

In this work, we propose a novel dynamic stealthy backdoor attack on SSL, called GhostEncoder, which can evade all existing defenses. GhostEncoder utilizes dynamic backdoor triggers that are specific to individual samples, resulting in diverse triggers for different backdoored samples. Importantly, perturbations are crafted to be globally imperceptible to human perception. Inspired by DNN-based image steganography Baluja (2017); Tancik et al. (2020); Zhu et al. (2018), we employ a pre-trained encoder-decoder network to encode an attacker-specified string into benign images. This encoding process generates sample-specific invisible additive noise, which serves as the backdoor trigger. The trigger is then seamlessly embedded into clean images, giving rise to backdoored images. Subsequently, the clean encoder undergoes fine-tuning to incorporate backdoors. This is achieved by bringing backdoored images closer to a reference input from the targeted class of a downstream task within the feature space. This enables the model to

establish a connection between the trigger and the targeted class, facilitating effective backdoor learning.

We conduct a comprehensive evaluation of GhostEncoder on two distinct pre-training datasets, spanning across three diverse downstream tasks. Experimental results demonstrate that GhostEncoder achieves a high attack success rate (ASR) while minimizing the impact on the accuracy of downstream classifiers. Notably, in a scenario where the clean image encoder is fine-tuned on CIFAR10 and the backdoor downstream classifier is constructed on STL10, GhostEncoder attains an impressive ASR of 96.57%. This success is attributed to the dynamic and unique trigger pattern introduced by GhostEncoder, which defies the underlying assumptions of certain existing defense mechanisms that anticipate the usage of fixed trigger patterns by attackers. As a result, GhostEncoder effortlessly circumvents such defenses, including our robust defense proposal, STRIP-CL. Furthermore, we investigate the effectiveness of GhostEncoder against two contemporary state-of-the-art defense approaches, namely DECREE Feng et al. (2023) and SSL-cleanse Zheng et al. (2023). While these defenses exhibit the capability to recover the trigger from backdoor images and offer a level of resilience against the attack, their success relies on the premise of a consistent trigger pattern employed by the attacker. In contrast, GhostEncoder's utilization of global, sample-specific perturbations renders these defenses ineffective in countering its dynamic attack strategy.

1.1. Contribution

Our key contributions are summarized as follows:

- We investigate the reasons for the failure of STRIP in self-supervised scenarios and propose a novel enhancement method called STRIP-CL. It improves the defense capabilities of the STRIP method and expands its defense scope, effectively countering backdoor attacks that use patch-type trigger patterns.
- We propose GhostEncoder, the first dynamic invisible backdoor attack on SSL, significantly enhancing the robustness of the attack.
- We systematically evaluate GhostEncoder on multiple upstream datasets and multiple downstream tasks to demonstrate the effectiveness of the attack.
- We explore three state-of-the-art defense mechanisms, to mitigate the impact of GhostEncoder. However, our experimental results emphasize the need for new defense approaches to effectively counter GhostEncoder.

2. Related work

2.1. Backdoor attack

2.1.1. Supervised learning backdoor techniques

The Backdoor attack is a rapidly growing and promising topic in the field of network security, which can have detrimental effects on security by poisoning the training process of DNNs. In the realm of supervised learning, researchers have extensively explored backdoor attacks targeted at image classifiers. These attacks involve the injection of patches, also known as triggers, into images. By augmenting clean datasets with toxic samples during training, attackers aim to manipulate the model's behavior. Notable works in this domain, including references Gu et al. (2017); Liu et al. (2018, 2017), emphasize the model's inclination to activate the backdoor effect exclusively on data containing the embedded triggers. Despite the compromised performance on clean data, models trained with poisoned datasets tend to misclassify images with triggers into the attacker's predefined target class during testing. While earlier studies, like Nguyen and Tran (2020), focus on sample-specific backdoor attacks, they often require meticulous control over training loss and data, diminishing their practical threat level. Moreover, the conspicuousness of trigger patterns, such as the visible 3x3 white square in Gu et al. (2017), renders them susceptible to defense mechanisms employed by state-of-the-art methods.

Concealing the trigger is pivotal for more sophisticated attacks. Chen et al. (2017) underline the importance of concealing triggers for enhanced stealthiness. They advocate for toxic samples that closely resemble benign ones, thereby making the attack more covert. A covert blending strategy, proposed by Chen et al., directly integrates backdoor triggers into benign images, avoiding blatant patching. Parallel studies explore covert attacks in various contexts. For instance, Quiring et al. (2020) address image scaling, while Saha et al. (2020) assume adversaries possess knowledge of the model structure. Authors in Zhang et al. (2022) exploit image structure to create covert trigger regions and embed them using a deep injection network, achieving concealed attacks. Similarly, in Wang et al. (2022), an RGB to YUV channel transformation combined with DCT conversion is proposed to insert triggers. These methods, while potent, maintain a fixed nature during training and testing.

2.1.2. Backdoor attack in SSL

The domain of SSL introduces distinct perspectives on backdoor attacks. Broadly, these attacks fall into two categories: poisoning-based Carlini and Terzis (2021); Saha et al. (2022); Liu et al. (2022); Li et al. (2023) and non-poisoning-based attacks Jia et al. (2022). Saha et al. (2022) introduced poisoning-based backdoor attacks within SSL, wherein adversaries contaminate a limited training dataset. These attacks yield fixed trigger patterns for backdoor samples, resulting in relatively modest success rates. Building on this, Li et al. (2023) proposed a novel poisoning-based attack approach, defining trigger patterns as specific frequency domain perturbations. Despite their potency, these trigger patterns remain static. Notably, backdoor attacks can manifest during various stages, including data collection, training, and deployment.

BadEncoder. Liu et al. (2022) devised a novel backdoor attack technique targeting pre-trained encoders within the realm of SSL. In computer vision, SSL primarily focuses on pre-training image encoders using extensive datasets of unlabeled images or image-text pairs. These encoders subsequently act as feature extractors for developing downstream classifiers that are optimized for various tasks requiring minimal or no labeled training data. BadEncoder represents an innovative approach by integrating backdoors into pre-trained image encoders. Defined by the authors as an optimization problem, it employs a specially designed objective function. This function correlates trigger patterns with reference inputs corresponding to the target

class within the representation space, thereby facilitating the misclassification of any triggered input into the target class in downstream tasks. This method boasts a high attack success rate coupled with a negligible reduction in accuracy. However, this approach still relies on a single, fixed trigger pattern, which can be successfully defended against by our proposed defense method - STRIP-CL.

2.2. Backdoor defense

Various defense methods have been proposed to counter backdoor attacks. Currently, defense efforts in the context of self-supervised scenarios primarily focus on detecting backdoor models or eliminating maliciously implanted Trojan horse models. To distinguish between backdoor and normal models, existing techniques employ vertical trigger patterns for model decision-making and discernment based on reverse trigger features (such as trigger size) Liu et al. (2019); Shen et al. (2021); Tao et al. (2022). DECREE Feng et al. (2023) belongs to the category of defense methods for detecting backdoor models. It is designed for backdoor detection in pre-trained encoders, requiring no classifier head or input labels, but it can only determine whether a model contains a backdoor, without the capability to remove it. Other methods utilize meta-classifiers to detect the presence of backdoors in models Kolouri et al. (2020); Feng et al. (2023); however, these methods entail significant computational resources and rely on a strong assumption—namely, that trigger size is known—thus rendering them less practical.

In the realm of removing tainted samples, Tran et al. (2018) attempt to cleanse toxic samples by analyzing the spectrum of latent features. Yet, this method assumes complete access to the infected training data, which is impractical in real-world applications. Additionally, Tejankar et al. (2023) take a novel approach by focusing on defense during the training phase of encoder models. They mitigate the impact of trigger patterns by searching for trigger-paste positions, effectively eliminating backdoors.

STRIP. Assuming that the backdoor trigger is input-agnostic, the defense proposed in Gao et al. (2019) aims to detect malicious samples before they are fed into the downstream classifier. To achieve this, the method filters suspicious images by superimposing various clean image patterns and observing the randomness of their prediction. The weight of the image superposition is equal for foreground and background images, and the randomness of the prediction is expressed through entropy. The smaller the entropy, the higher the probability that a suspicious image is malicious.

SSL-Cleanse. Recently, Zheng et al. (2023) introduced a new defense method, SSL-Cleanse. It employs clustering algorithms and reverse identification techniques for effective backdoor detection. By clustering encoder-generated samples, SSL-Cleanse pinpoints clusters with anomalous trigger patterns. It then applies reverse pattern recognition to identify and remove these triggers. Significantly, SSL-Cleanse is adept at recognizing and mitigating training-agnostic attacks, like Trojan threats, in SSL encoders, maintaining robust performance even in the absence of downstream task labels. This approach introduces an innovative defense against possible backdoor vulnerabilities in SSL encoders.

2.3. Applications of self-supervised learning

Modern computer vision systems excel in various challenging computer vision tasks, largely relying on the availability of large annotated datasets. However, acquiring such data is both time-consuming and expensive, making it impractical to construct large-scale labeled datasets in many real-world scenarios. Therefore, devising learning methods that can successfully recognize new concepts with only a small number of labeled examples is a crucial research challenge.

SSL is a general framework that aims to train image encoders from large amounts of unlabeled data, extracting useful information directly

from the data itself, unlike supervised learning which requires manually labeled data. Among many methods, such as MoCo He et al. (2020b), SimCLR Chen et al. (2020a), Sim-CLRV2 Chen et al. (2020b), and CLIP Radford et al. (2021), contrastive learning achieves state-of-the-art performance. This method compares samples with semantically similar examples (positive pairs) and those with semantically dissimilar ones (negative pairs). By designing the model structure and comparison loss, the positive pairs with similar semantics are represented closer in the representation space, while the semantically dissimilar negative pairs correspond to farther representations, achieving a similar clustering effect.

In the field of computer vision, SSL has numerous applications. Doersch et al. Doersch et al. (2015) proposed a method for training a convolutional neural network (CNN) by predicting the relative position of two randomly sampled, non-overlapping image patches. Later, papers in Noroozi and Favaro (2016); Noroozi et al. (2018) extended this idea to predicting the permutation relationship between multiple randomly sampled and permuted blocks. Besides patch-based methods, some techniques utilize image-level self-supervised losses. For instance, in Zhang et al. (2016), authors proposed using the colorization of grayscale images as an auxiliary task. Another example of a pretext task is in Komodakis and Gidaris (2018), which predicts the angle of the rotation transformation applied to the input image.

3. Preliminaries

3.1. Image steganography

Image steganography is a sophisticated method that involves concealing confidential data within an image, aiming to maintain its imperceptibility to observers while minimizing any discernible alterations to the original image. The concealable amount of information is typically quantified in bits per pixel (bpp), where a lower bpp value signifies a reduced level of concealed data to mitigate the risk of detection. However, despite these efforts, advanced statistical analyses and detection techniques can potentially uncover these covert messages. In response to this challenge, the research focus has shifted toward leveraging Deep Neural Networks (DNNs) as a solution. DNNs offer heightened embedding capacity and enhanced security measures to counteract these vulnerabilities.

In their work, the authors propose a new method for image steganography using DNNs Baluja (2017). Their methodology entails the training of two interconnected networks: a hiding network and a revealing network, which collectively facilitate the embedding and extraction of full-size color images within other images of equivalent dimensions. The hiding network operates by taking a cover image and a transformed secret image as input, subsequently generating a container image. To prepare the secret image for processing, a pre-network operation is performed, which involves resizing the image if needed and converting its color-based pixels into more suitable feature representations, such as edges. Conversely, the revealing network is responsible for decoding the container image, unveiling the concealed secret image.

During the training process, the authors minimized the following error terms between the cover and secret images for reconstruction:

$$\mathcal{L}(\mathbf{c}, \mathbf{c}', \mathbf{s}, \mathbf{s}') = \|\mathbf{c} - \mathbf{c}'\| + \beta \|\mathbf{s} - \mathbf{s}'\| \quad (1)$$

where $\mathbf{c}$ and $\mathbf{s}$ represent the cover and secret images, $\mathbf{c}'$ and $\mathbf{s}'$ represent the reconstructed images, and β is the weight that balances the importance of each error term. It is worth noting that the error term $\|\mathbf{c} - \mathbf{c}'\|$ does not apply to the weights of the Reveal Network, ensuring that the information of the container image does not affect the extraction of the secret image.

To prevent the network from relying solely on LSBs, a small amount of noise is added to the container image during the training process. This noise occasionally flips LSBs, preventing them from being the

exclusive carriers for reconstructing the secret image. This method extends the principles of traditional steganography methods by leveraging the capabilities of DNNs to distribute the secret image's representation across multiple bits of the carrier image. By doing so, this approach transcends the constraints of conventional LSBs operations and achieves a greater embedding capacity.

3.2. Self-supervised learning

SSL has become a popular approach in the field of machine learning and computer vision because it allows learning representations from unlabeled data. By learning to predict certain attributes or relationships in the data, the model becomes capable of capturing valuable and meaningful representations that can subsequently be leveraged for various downstream tasks. The SSL pipeline typically consists of two stages: pre-training an image encoder and constructing a downstream classifier, which is then fine-tuned.

Among many methods, contrastive learning has demonstrated remarkable performance, such as MoCo He et al. (2020b), SimCLR Chen et al. (2020a), SimCLRV2 Chen et al. (2020b), and CLIP Radford et al. (2021). Contrastive learning achieves this by contrasting samples with their semantically similar (positive) and dissimilar (negative) counterparts. Through meticulous design of the model architecture and contrastive loss function, the method ensures that the representations of positively related pairs are drawn closer in the representation space, while the representations of negatively related pairs are pushed farther apart, thereby yielding clustering-like effects. Notably, SimCLR, introduced by Chen et al. in 2020 Chen et al. (2020a), is a prominent approach that employs contrastive learning to acquire robust visual representations devoid of manual annotation. Its primary objective is to maximize the coherence between various perspectives of the same sample, while concurrently minimizing the coherence between different samples' perspectives.

SimCLR adopts a two-step training process, including data augmentation and contrastive loss optimization. In the first step, SimCLR applies strong data augmentation techniques such as random cropping, color distortion, and Gaussian blur to create different views of each input sample. These augmented views are beneficial for learning. In the second step, SimCLR leverages a contrastive loss function to guide the model in bringing positive representations closer while simultaneously pushing negative representations further apart. The calculation of the contrastive loss hinges on the cosine similarity between the representations of the augmented views. The overarching objective is to maximize the similarity between positive pairs while minimizing the similarity between negative pairs.

The contrastive loss formula used in SimCLR is as follows:

$$\ell_{i,j} = -\log \left(\frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j) / \tau)}{\sum_{k=1}^{2N} \mathbb{I}(k \neq i) \cdot \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k) / \tau)} \right) \quad (2)$$

where $\mathbb{I}(k \neq i) \in \{0, 1\}$ is an indicator function evaluating to 1 iff $k \neq i$. $\mathbf{z}_i$ and $\mathbf{z}_j$ represent positive pairs, $\mathbf{z}_k$ represents negative pairs, and τ is a temperature parameter that controls the sharpness of the distribution.

SimCLR has shown promising results in various computer vision tasks, including image classification, object detection, and image segmentation. Through its adeptness at acquiring comprehensive and distinctive visual representations, SimCLR has established itself as a leader in the field. This is evidenced by its attainment of state-of-the-art performance on benchmark datasets, underscoring its capacity to generate impactful and discriminative visual features.

4. Methodology

Our proposed defense and attack are as follows. The definition of the notations is presented in Table 1.

Table 1
A summary of notations.

Name	Notation	Meaning
Clean Encoder	e	The target encoder used to inject the backdoor
Backdoored Encoder	e'	An encoder that has been injected with a backdoor
(Target downstream task, target class)	(T_i, c_i)	A pair of target downstream tasks and target classes
Reference Inputs	R_i	one or more images in the target class for each (target downstream task, target class) pair
A Normal Sample	x	A clean input image
A Backdoor Trigger	p_x	A disturbance that triggers a back door
A Backdoor Sample	$x \oplus p_x$	A backdoor instance as the model input
Shadow Dataset	D_s	A collection of backdoor images
Hyper-parameter	λ_1	Parameters used by the backdoor injection process
Steganographic encoder	e_i	An encoder that adds a watermark to a clean image

4.1. Threat model

We only consider backdoor defense and attack on visual encoders.

Defender's Goals. The defender's goal is to detect and mitigate backdoor samples while minimizing the impact on the overall model performance.

Defender's Capacities. We assume that the defender has access to the suspected encoder and obtains feature vectors of input images from the downstream dataset to train the downstream classifier. Operating under the constraints of not knowing the trigger or target categories and lacking access to reliable data or image labels, the defender faces the challenge of defending against these attacks. Our primary focus is on investigating backdoor attacks that utilize a static patch-type trigger pattern.

Attacker's Goals. In general, backdoor attackers intend to embed hidden backdoors in DNNs through data poisoning. The hidden backdoor will be activated by the attacker-specified trigger, i.e., the prediction of the image containing the trigger will be the target label, no matter what its ground-truth label is. In particular, attackers have three main goals, effectiveness, stealthiness, and sustainability. The effectiveness requires that the prediction of attacked DNNs should be the target label when the backdoor trigger appears, and the performance on benign testing samples will not be significantly reduced. The stealthiness requires that adopted triggers should be concealed and invisible to the human eye. The sustainability requires that the attack should still be effective under some common backdoor defenses.

Attacker's Capacities. Our work considers two possible attackers: 1) Untrusted service providers who may inject backdoors into their pre-trained image encoders and provide them to downstream users, 2) Malicious third parties who inject backdoors into image encoders pre-trained by service providers and publish them on the internet for downstream users to download and use. The attacker has access to the watermark dataset used for training the watermark generator, clean pre-trained image encoders, unlabeled images (referred to as shadow dataset), and some images for each pair (target downstream task, target class) known as reference inputs. However, the attacker cannot access the downstream dataset used to construct the downstream classifiers or influence the training process of the downstream classifiers.

4.2. Detecting static backdoor attack without pre-training dataset

Existing solutions struggle to detect backdoors in encoders pre-trained in a self-supervised setting, as they are designed for supervised learning that requires classification labels. We propose an enhanced defense, STRIP-CL, building upon STRIP. We demonstrate the defensive effectiveness of both methods against BadEncoder Jia et al. (2022) and SSL-Backdoor Saha et al. (2022). Additionally, we analyze the reasons for STRIP's ineffectiveness.

4.2.1. Algorithm overview of STRIP-CL

However, existing solutions designed for supervised learning face challenges when detecting backdoors in pre-trained encoders since they

require classification labels. To address this limitation, we propose an enhanced version of the STRIP, called STRIP-CL, tailored specifically for contrastive learning. As described in Section 2.2, when using an encoder with strong feature extraction capabilities, the superimposition of image mode in STRIP has no noticeable effect on clean images. As a result, the entropy distributions of clean samples and malicious samples overlap, making them difficult to distinguish. STRIP-CL observes the maximum prediction distribution to determine if an input image is malicious. Specifically, we generate a distribution map for STRIP-CL using hard labels, counting the prediction outcomes from the input image classifier after superimposing multiple clean images onto each input image. The highest count predicted for a certain class serves as the abscissa (x), while the ratio of samples with the maximum count of x to the total is taken as the ordinate (y). We classify an image as malicious based on the abscissa threshold. In comparison to STRIP, our proposed method is better suited for detecting sample-agnostic triggers in self-supervised scenarios.

4.2.2. Defense assessment

Fig. 2 illustrates the results of the STRIP and STRIP-CL defenses against BadEncoder and SSL-Backdoor. The entropy distribution of these methods is showcased in Fig. 2a and 2c. In these graphs, the horizontal axis signifies the normalized entropy values, while the vertical axis denotes the corresponding percentage of inputs with these entropy values. Moreover, Fig. 2b and 2d present the maximum prediction distribution, where the horizontal axis (x) indicates the highest frequency of predictions for a given class, and the vertical axis (y) reflects the proportion of samples with this maximal prediction frequency relative to the overall sample population. In this experiment, both BadEncoder and SSL-Backdoor utilize an upstream dataset of CIFAR10 and a downstream dataset of STL10 to construct the target model. To enhance the success rate of SSL-Backdoor, which is typically around 33.1% Zheng et al. (2023), we increase its poisoning rate to 5%. After achieving an elevated attack success rate of 82%, we apply the defense method.

From Fig. 2a and 2c, it can be observed that the original STRIP fails to defend against BadEncoder and SSL-Backdoor attacks, as the distribution of the backdoored samples generated by these two attacks overlaps with that of the clean samples, making it challenging to distinguish them through a simple threshold setting. Conversely, Fig. 2b and 2d demonstrate that clean samples tend to be located on the left side of the distribution map, closely following a normal distribution, while malicious samples predominantly occupy the far right side of the distribution map. Therefore, STRIP-CL can effectively filter out and remove the backdoor samples from the backdoored training set of BadEncoder and SSL-Backdoor, significantly reducing their attack success rates. We found it necessary to adjust the weights applied in the image overlay appropriately. Here, we chose a weight value of 0.9, as this setting leads to more distinct classification boundaries in the distribution map.

4.3. How to generate trigger

Inspired by DNN-based image steganography, ISSBA Li et al. (2021) encodes a specified string into benign images as backdoor images using

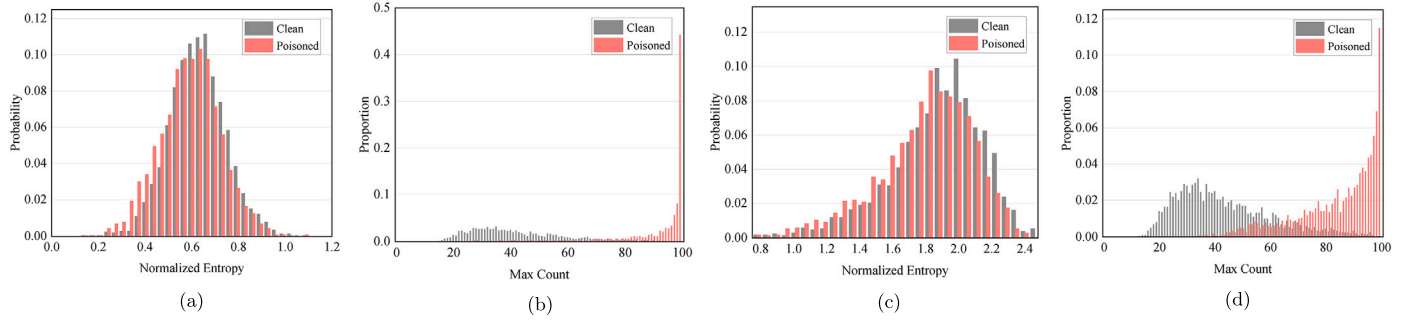


Fig. 2. Maximum prediction distribution map for detecting BadEncoder and SSL-Backdoor attack, Strip and Strip-Cl. (a) Maximum predicted distribution of Strip defense BadEncoder; (b) Maximum predicted distribution of Strip-Cl defense BadEncoder; (c) Maximum predicted distribution of Strip defense SSL-Backdoor; (d) Maximum predicted distribution of Strip-Cl defense SSL-Backdoor.

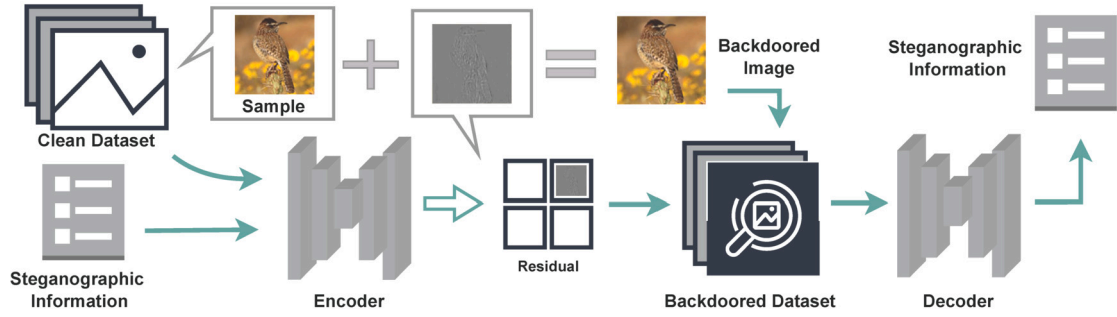


Fig. 3. The training process of the encoder-decoder network.

a pre-trained encoder. However, this method operates in a supervised scenario, requiring more annotated data and utilizing target label information as the content of the steganographic string. In contrast, our method is implemented in a self-supervised setting, freeing itself from label constraints and not limiting the specific content of the string. It only requires consistent encoding information in the backdoored images.

Generating Sample-Specific Triggers. We exemplify with a pre-trained encoder-decoder network to generate sample-specific triggers. The watermark encoder's output varies based on the input image, yielding invisible additional noise as sample-specific triggers. As depicted in Fig. 3, we create the backdoored dataset by adding watermarks to the clean dataset. The encoder first takes a benign image and watermark information to generate a residual image (i.e., the corresponding trigger). Overlaying this residual image onto the benign image yields the encoded image. The watermark encoder and decoder are jointly trained on the benign training set (watermark dataset). Throughout the training, a uniform string functions as steganographic content, training the watermark encoder to embed it into images, simultaneously minimizing the perceptual difference between input and encoded images. The decoder, on the other hand, learns to recover hidden messages from encoded images. Using this well-trained watermark encoder, we append sample-specific triggers to input images, generating backdoor images.

4.4. Stealthy backdoor attacks with dynamic triggers to pre-trained encoders

In this section, we will elucidate our proposed method, GhostEncoder. Before delving into the generation of sample-specific triggers, we provide a concise overview of the GhostEncoder procedure. Our approach is formulated as the following optimization problem, like Jia et al. (2022).

Formally, to achieve the goals of effectiveness and practicality mentioned earlier, we propose modifying the clean image encoder e to obtain a backdoor encoder e' . For each (target downstream task, target class) pair (T_i, C_i) , the attacker selects a set of reference inputs

$\mathcal{R}_i = \{x_{i1}, x_{i2}, \dots, x_{ir_i}\}$ from the target class C_i , where p_x is the trigger generated for each input x . $x \oplus p_x$ is the backdoor input. We propose using efficiency loss and utility loss to quantify the efficiency and utility objectives, respectively. Our effectiveness loss consists of the following two formulas:

$$L_0 = - \frac{\sum_{i=1}^t \sum_{j=1}^{r_i} \sum_{x \in \mathcal{D}_s} s(e'(x \oplus p_x), e'(x_{ij}))}{|\mathcal{D}_s| \cdot \sum_{i=1}^t r_i} \quad (3)$$

$$L_1 = - \frac{\sum_{i=1}^t \sum_{j=1}^{r_i} s(e'(x_{ij}), e(x_{ij}))}{\sum_{i=1}^t r_i} \quad (4)$$

where $s(\cdot)$ measures the similarity between two feature vectors, with higher similarities resulting in increased s values and reduced L_0 and L_1 values. $|\mathcal{D}_s|$ represents the number of inputs in the shadow dataset, and the denominators within L_0 and L_1 serve to normalize the loss values. Our effectiveness loss is a weighted sum of the two terms $L_0 + \lambda_1 \cdot L_1$, where λ_1 is a hyperparameter to balance the two terms. A smaller L_0 indicates that the backdoor image encoder e' generates closely resembling feature vectors for both reference and backdoor inputs in the shadow dataset. On the other hand, a smaller L_1 suggests similar feature vector production by both the backdoor and clean image encoders for the reference inputs, ensuring that the downstream classifier, influenced by the backdoor, accurately predicts the target class based on L_0 .

The main goal of the utility loss is to preserve the accuracy of downstream classifiers that rely on our backdoored image encoder for clean inputs. This requires ensuring that the backdoor image encoder and the clean image encoder generate similar feature vectors for clean inputs, thereby achieving a smaller utility loss. To achieve this goal, we define a utility loss:

$$L_2 = - \frac{1}{|\mathcal{D}_s|} \cdot \sum_{x \in \mathcal{D}_s} s(e'(x), e(x)) \quad (5)$$

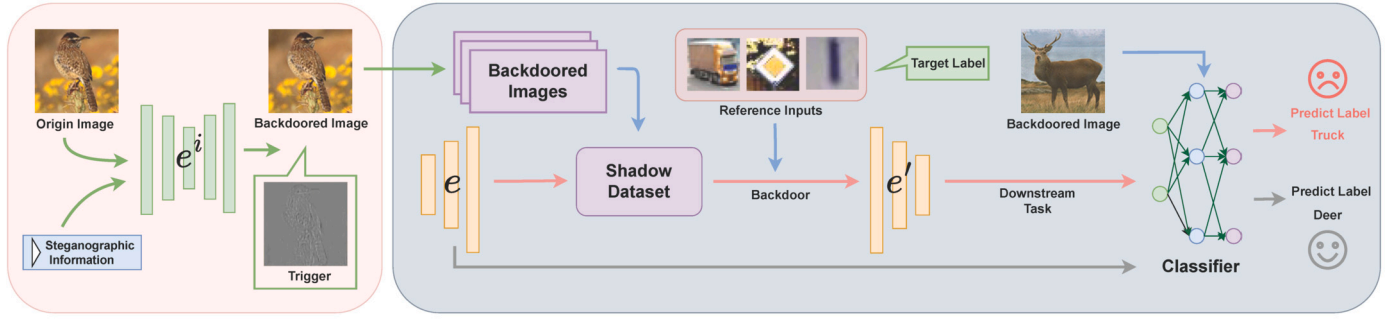


Fig. 4. An overview of proposed GhostEncoder.

After defining the three loss terms L_0 , L_1 and L_2 , we tackle the optimization problem of the backdoor image encoder e' based on these terms.

$$\min_{\theta} L = L_0 + \lambda_1 \cdot L_1 + \lambda_2 \cdot L_2 \quad (6)$$

The two hyperparameters λ_1 , λ_2 are employed to balance these loss terms. Their impact on the backdoor encoder will be thoroughly examined during the evaluation process.

Pipeline: In this work, we propose a novel dynamic invisible backdoor attack in the context of SSL, named GhostEncoder. Fig. 4 provides an overview of GhostEncoder. The left part illustrates the generation stage of backdoored images. Here, we employ an additional watermark encoder, denoted as e' , to embed hidden information into the backdoor images, generating the triggers. This encoder is derived from a pre-trained encoder-decoder network using a watermark dataset. On the right side of Fig. 4, we present the stages of backdoor injection and downstream task inference. The obtained embedded triggers in the backdoor images serve as the shadow dataset, while the reference inputs consist of images collected from the target class on the web. The objective of the backdoor injection phase is to construct a backdoor image encoder e' from a clean image encoder e , achieving the goals of effectiveness, invisibility, and robustness. During the inference stage, the downstream classifier built upon our backdoor image encoder will predict any input containing the corresponding trigger to be of the target class, without affecting the normal performance of non-target downstream tasks.

5. Evaluation

5.1. Experimental setup

5.1.1. Datasets

In this work, we employ four distinct image datasets, namely CIFAR10, STL10, GTSRB, and SVHN, to facilitate comprehensive evaluation.

- **CIFAR10:** This dataset comprises 50,000 training images and 10,000 testing images, with each image sized at 32x32x3 pixels. The images are categorized into ten different classes.
- **STL10:** Consisting of 5,000 labeled training images and 8,000 labeled testing images, the STL10 dataset includes images sized at 96x96x3 pixels. It encompasses ten distinct classes, and notably, there are an additional 100,000 unlabeled images available.
- **GTSRB:** Comprising 51,800 traffic sign images distributed across 43 categories, the GTSRB dataset is divided into 39,200 training images and 12,600 testing images.
- **SVHN:** With each image representing a digit from Google Street View's house number dataset, the SVHN dataset contains 73,257 training images and 26,032 testing images, all sized at 32x32x3 pixels. Additionally, the dataset is characterized by some scattered digits near the digits of interest, rendering it a noisy dataset.

5.1.2. Parameter setting

In our experiments, we employed CIFAR-10 or STL-10 as the pre-training datasets due to their larger image capacity and non-noisy nature. For both of these pre-training datasets, we selected the ResNet-18 He et al. (2016) architecture as the framework for our image encoders. During training, we utilized SimCLR Chen et al. (2020a) and applied a collection of augmentation operators, including random resizing-cropping, horizontal flipping, color distortion, and random grayscale. We employed the Adam optimizer and conducted training for 1000 epochs to obtain the image encoders. The implementation of this process was based on publicly available code from SimCLR.^{1,2}

The triggers used in our approach are generated by watermark encoders trained on watermark datasets. The watermark encoder is trained using an encoder-decoder network configuration consistent with StegaStamp Tancik et al. (2020). In this configuration, the encoder network adopts a U-Net Ronneberger et al. (2015) style, and the decoder network employs a spatial transformer network Jaderberg et al. (2015). The training incorporates four loss terms: L2 residual regularization, LPIPS perceptual loss Zhang et al. (2018) to minimize perceptual distortion, cross-entropy loss for concealed information reconstruction, and perceptual distortion loss. Scaling factors for these losses are set to 2.0, 1.5, 0.5, and 1.5, respectively. During training, we employed both the Adam Kingma and Ba Adam (2014) optimizer and the SGD optimizer, with initial learning rates set to 0.0001 and 0.001, respectively. The learning rate is decayed by a factor of 0.1 after the 15th and 20th epochs. The resulting encoder is utilized to create backdoor images for generating a shadow dataset, consisting of 50,000 images extracted from the pre-training dataset.

The Backdoor injection phase begins with the assumption that the attacker selects a target downstream task/dataset and a target category. When the upstream dataset is CIFAR-10, we employ STL-10, GTSRB, and SVHN as downstream datasets. In the case of an upstream dataset of STL-10, we opt for CIFAR-10, GTSRB, and SVHN as downstream datasets. Additionally, "airplane," "truck," "priority sign," and "digit one" are chosen as target categories for CIFAR-10, STL-10, GTSRB, and SVHN datasets, respectively. We assume a reference input for each (target downstream task, target category) pair, collected by correctly classifying acquired reference inputs as the target category using the backdoored downstream classifier. We perform fine-tuning on the pre-trained image encoder with a learning rate of 0.001 and a batch size of 256 for 200 epochs to inject the backdoor unless otherwise specified. Parameters λ_1 and λ_2 default to 1 unless specified otherwise.

Given an image encoder (with a backdoor) pre-trained using a pre-training dataset, we utilize it as the feature extractor for downstream tasks. For a chosen dataset as the downstream dataset, its training data train the downstream classifier, while its testing data evaluate the classifier. Specifically, we employ a two-layer Multilayer Perceptron (MLP) with a hidden layer size of 128 as the downstream classifier, employ-

¹ <https://github.com/google-research/simclr>.

² <https://github.com/leftthomas/SimCLR>.

Table 2
Attack effect comparison between BadEncoder and GhostEncoder.

Pre-training Dataset	Method	Downstream Dataset	CA (%)	BA (%)	ASR-B (%)	ASR (%)	ASR-after Defense(%)
CIFAR10	BadEncoder	STL10	76.14	76.18	10.37	99.73	15.8 (83.93 ↓)
		GTSRB	81.84	82.27	5.09	98.64	14.5 (84.14 ↓)
		SVHN	61.72	69.32	35.17	99.14	3.2 (95.94 ↓)
	GhostEncoder	STL10	76.14	76.28	8.61	96.57	95.30 (1.27 ↓)
		GTSRB	81.84	64.92	4.83	65.19	65.05 (0.14 ↓)
		SVHN	61.72	55.98	20.98	24.41	24.05 (0.36 ↓)
STL10	BadEncoder	CIFAR10	87.36	86.82	10.13	94.35	21.0 (73.35 ↓)
		GTSRB	76.83	78.34	4.86	88.36	19.88 (68.48 ↓)
		SVHN	54.72	62.91	34.48	23.37	21.45 (1.92 ↓)
	GhostEncoder	CIFAR10	87.36	84.97	10.02	90.28	87.82 (2.46 ↓)
		GTSRB	76.83	70.54	3.68	67.16	66.26 (0.90 ↓)
		SVHN	54.72	50.62	19.18	27.82	26.44 (1.38 ↓)

ing the cross-entropy loss function and Adam optimizer during training. Furthermore, the downstream classifier is trained for 500 epochs with an initial learning rate of 0.0001. Unless stated otherwise, CIFAR-10 serves as the default pre-training dataset, and STL-10 is the default downstream dataset. Note that we resize each image in the STL-10 dataset to a size of $32 \times 32 \times 3$ to maintain consistency with other datasets.

5.1.3. Evaluation metrics

To effectively assess the performance of the attacks, we primarily employ three metrics, namely Clean Accuracy (CA), Backdoored Accuracy (BA), and Attack Success Rate (ASR), to evaluate our GhostEncoder targeting the classification model. Here, CA and BA represent the classification accuracy of downstream classifiers constructed based on clean and backdoored encoders, respectively, on the downstream clean test dataset. ASR indicates the proportion of trigger-embedded samples in which the downstream classifier built upon the backdoored encoder predicts the corresponding target class.

Furthermore, we introduce two additional metrics for supplementary evaluation: ASR-B and Attack Success Rate after Defense (ASR-AD). ASR-B measures the percentage of trigger-embedded backdoor images recognized as the target class by the downstream classifier constructed with a clean encoder, reflecting the success rate of attacks when the attacker has not injected a backdoor into the pre-trained image encoder. To assess the robustness of the defense method against attacks, we introduce the ASR-AD. It quantifies the proportion of test inputs with embedded triggers accurately predicted as the target class by the backdoored downstream classifier after undergoing the Strip-Cl defense strategy.

5.2. Effectiveness and efficiency of attack

As shown in Table 2, we evaluated BadEncoder and GhostEncoder on three different downstream datasets, each with two pretraining datasets. We provided the CA and ASR-B for the pre-trained image encoders without injecting backdoors, as well as the BA, ASR, and ASR-AD for GhostEncoder. Overall, our GhostEncoder achieved the following:

High attack success rate without compromising model utility: Through experimentation, we discovered that the effectiveness of downstream attacks is influenced by the strength of the watermark. Depending on the distribution of the downstream dataset, the closer the distribution of the downstream dataset is to the pre-training dataset, the stronger the watermark is required. Consequently, we trained a robust watermark encoder for STL-10 and a less potent watermark encoder for GTSRB and SVHN, respectively.

Based on the results obtained from using two different pretraining datasets on three different downstream datasets, GhostEncoder demonstrated a high ASR. For instance, when the pretraining dataset was CIFAR10 and the downstream dataset was STL10, GhostEncoder achieved an ASR of 96.57%. In contrast, without injecting a backdoor into the

pre-trained image encoder, the ASR-B was only 8.61%. The decrease in BA relative to CA was only 0.14% (CA: 76.14%, BA: 76.28%). Thus, our GhostEncoder maintained the accuracy of the downstream classifier while achieving a high ASR. We noted that the ASR-B was relatively higher when the downstream dataset was SVHN due to its class imbalance, with the selected target class being the most popular category.

Resistance against state-of-the-art defenses: ASR-AD represents the results of applying the STRIP-Cl method (introduced in Section 4.2.1) for defense. It involves setting a threshold to filter out toxic datasets, resulting in the filtered ASR-AD. As described in Section 4.2.1, the threshold was determined by selecting the maximum count of hard-label predictions for a specific class as the x-coordinate and setting a threshold to filter out toxic samples. According to the results in the ASR-AD column of the table, values in parentheses indicate the threshold. These results demonstrate that our GhostEncoder can resist STRIP-Cl defense. For example, when the pretraining dataset was CIFAR10 and the downstream dataset was GTSRB, GhostEncoder achieved a defense success rate of 65.05%, compared to an ASR of 65.19% without defense, resulting in a mere 0.14% decrease in ASR. Importantly, in most cases, the difference between the ASR and the defense success rate was within 0.5%. Thus, STRIP-Cl had minimal impact on weakening GhostEncoder's attack effectiveness. In contrast, the ASR of BadEncoder significantly decreased after the defense. For instance, when the pretraining dataset was CIFAR10 and the downstream dataset was GTSRB, its ASR dropped from 98.64% to 14.5%, rendering the attack nearly ineffective. In most cases, STRIP-Cl was able to completely defend against BadEncoder.

5.3. Hyperparameter analysis

5.3.1. Impact of shadow dataset

The shadow dataset can be characterized by its size and distribution. Therefore, we investigate the impact of the shadow dataset size on GhostEncoder. Fig. 5a illustrates the influence of shadow dataset size on GhostEncoder. Taking STL10 as the target downstream dataset, we observe that when the shadow dataset size is around 25% of the pre-training dataset, GhostEncoder achieves an ASR of over 60%. Once the shadow dataset size surpasses about 50% of the pre-training dataset, GhostEncoder's ASR reaches as high as 96%, nearly approaching the performance achieved when the shadow dataset size equals the pre-training dataset, while maintaining the accuracy of the downstream classifier. This suggests that our injected dataset for the backdoor requires only a minimal amount of information from the pre-training dataset.

5.3.2. Impact of other parameters

Our GhostEncoder utilizes a loss function described by Equation (5), which consists of three loss terms: L_0 , L_1 , and L_2 . We use two hyperparameters, λ_1 and λ_2 , to compute the weighted loss. L_1 serves the effectiveness objective, achieving a high success rate in backdoor

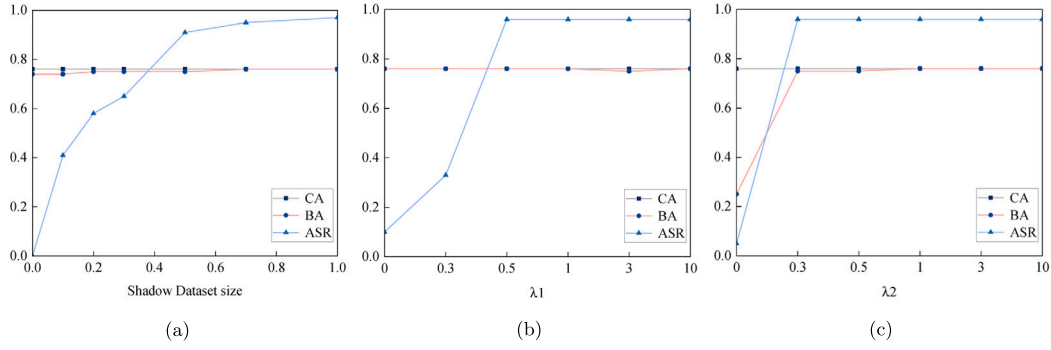


Fig. 5. The impact of the various parameters on BadEncoder when the pre-training dataset is cifar10 and the target downstream dataset is STL10. (a) The impact of the shadow dataset size (b) The impacts of λ_1 (c) The impacts of λ_2 .

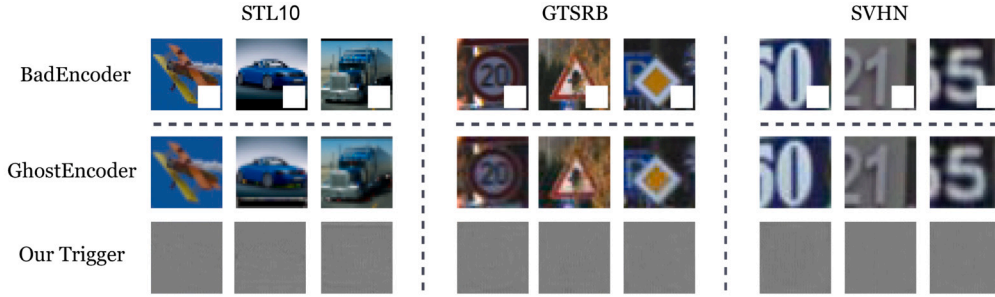


Fig. 6. Comparison of the backdoor samples of BadEncoder and GhostEncoder. The third line is the trigger generated by the GhostEncoder attack.

attacks, while L_2 serves the utility objective, maintaining high accuracy after significant perturbation, thus preserving the accuracy of the downstream classifier. By controlling the parameters λ_1 and λ_2 , we can balance the accuracy of perturbation and the success rate of the attack. Therefore, we investigated the influence of λ_1 and λ_2 on our GhostEncoder. Fig. 5b ($\lambda_2 = 1$) and Fig. 5c ($\lambda_1 = 1$) illustrate the results.

Initially, we observe that GhostEncoder achieves high attack success rates and maintains classification accuracy when λ_1 and λ_2 exceed certain thresholds. For instance, when λ_1 is set to 0.5, GhostEncoder already achieves a high ASR of 96.18%. Furthermore, GhostEncoder exhibits higher sensitivity to smaller numerical ranges of λ_1 , while for larger numerical values of λ_1 , even when reaching 10, the attack success rate remains stable. This stability arises from the relatively small training set size of the STL10 dataset. The distribution of λ_2 is similar to that of λ_1 , showing heightened sensitivity to smaller numerical ranges and maintaining a consistent ASR without fluctuations or declines in larger numerical ranges. We attribute this behavior to the shadow dataset's size (related to L_2), which significantly surpasses the quantity of reference inputs.

5.4. Attack stealthiness

Fig. 6 presents the backdoored images generated using the BadEncoder and our GhostEncoder attack methods. Here, we showcase the backdoored images for three downstream datasets: STL10, GTSRB, and SVHN, along with the trigger images utilized in our proposed GhostEncoder method. The BadEncoder's trigger consists of a fixed 10×10 white square, positioned at the bottom right corner of the image. This trigger is singular, fixed, and clearly visible, lacking effective concealment. In contrast, the trigger employed in the backdoored images of our GhostEncoder method is sample-specific, exhibiting a remarkably high level of concealment, and imperceptible to the human eye.

To assess the concealment efficacy of the embedded trigger, we performed extensive experiments evaluating the distinctions between clean and backdoor-introduced samples with respect to feature representation and input data. Our evaluation focused on the perceptibility of

Table 3

Invisibility of the triggering pattern used by GhostEncoder. Larger PSNR and SSIM, and smaller LPIPS are better.

Method	PSNR	SSIM	LPIPS
No Attack	INF	1.000	0.000
CIFAR10	33.896	0.985	0.022
STL10	33.828	0.974	0.024
GTSRB	32.873	0.980	0.023
SVHN	37.971	0.996	0.009

the corrupted images to the human eye, utilizing established similarity indices such as peak signal-to-noise ratio (PSNR) Huynh-Thu and Ghanbari (2008), structural similarity index (SSIM) Wang et al. (2004), and learned perceptual image patch similarity (LPIPS) Zhang et al. (2018). PSNR and SSIM are predicated on pixel-level statistical comparisons, whereas LPIPS leverages feature similarities discerned by a pretrained AlexNet model. Elevated SSIM and PSNR readings correlate with a higher visual indistinguishability between the backdoored and original images, implying a more effective concealment of the backdoor samples. In contrast, lower LPIPS scores denote a higher degree of stealth, indicating the trigger's successful obfuscation.

The combined analysis of Fig. 6 and Table 3 reveals that the trigger employed by BadEncoder is conspicuously visible. In contrast, as delineated in Table 3, we have evaluated the trigger pattern of GhostEncoder against all three fidelity metrics. This evaluation substantiates the trigger pattern's invisibility utilized by GhostEncoder.

5.5. Robustness to defense

In this work, we investigate two emerging defenses for SSL: our proposed Strip-Cl and SSL-Cleanse. Through experiments, we demonstrate that our GhostEncoder attack can successfully bypass both of these defenses.

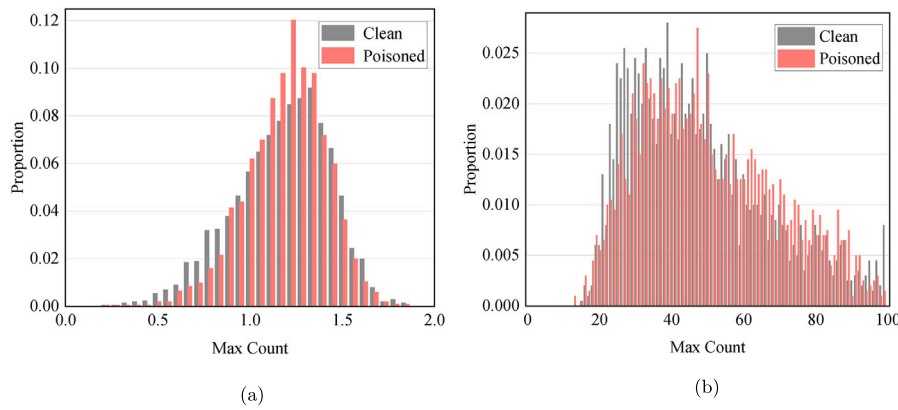


Fig. 7. Maximum prediction distribution map for detecting GhostEncoder, Strip and Strip-Cl. (a) Maximum predicted distribution of Strip defense GhostEncoder; (b) Maximum predicted distribution of Strip-Cl defense GhostEncoder.

Table 4

The certified accuracy and attack success rates for the backdoored downstream classifiers defended by SSL-Cleanse Zheng et al. (2023). The pre-training dataset is CIFAR10.

Downstream Dataset	CA	BA	ASR
STL10	76.14	24.05	7.75
GTSRB	81.84	5.90	0.00
SVHN	61.72	52.11	24.13

5.5.1. Defense of Strip-Cl

As shown in Fig. 7, Fig. 7a and 7b illustrate the maximum prediction maps obtained by the original STRIP method and our proposed STRIP-Cl method for defending against GhostEncoder with hard labels. From the figure, it can be observed that regardless of using STRIP or STRIP-Cl defense, the distributions of backdoored and clean samples generated by GhostEncoder are highly overlapping. Distinguishing backdoored samples from clean ones through simple thresholding becomes challenging. Therefore, both methods fail to defend against GhostEncoder effectively.

Regarding patch-based perturbation approaches, including data poisoning and model weight modification types of backdoor attacks, one common characteristic is the use of a single and fixed trigger, which contributes to their robustness. Consequently, when overlaying randomly selected clean images onto backdoored images, the trigger remains effective in the overlaid image, resulting in a strong classification bias. The confidence vector of soft labels tends to favor the target label and exhibits reduced ambiguity. Conversely, for global and sample-specific trigger patterns, the triggers lack the characteristic of being single and fixed. Due to this limitation in the STRIP method's concept, the enhanced method STRIP-Cl can only detect patch-based perturbations and fails to identify global perturbation types.

5.5.2. Defense of SSL-Cleanse

SSL-Cleanse utilizes clustering algorithms and reverse recognition techniques to detect backdoor attacks. By clustering samples generated by the SSL encoder, it can identify groups of samples with anomalous triggers. Then, through reverse pattern recognition techniques, SSL-Cleanse can identify and remove these triggers, restoring the encoder to a clean state without backdoors.

However, for the GhostEncoder attack proposed in this work, which utilizes global perturbations that affect the entire image space, the trigger has an impact on every pixel, unlike patch triggers that are limited to specific locations. This makes the detection and removal of triggers more challenging, and the sample-specific nature of GhostEncoder makes it difficult to recognize the trigger as a unified pattern, further hindering direct identification and elimination.

In this work, we assume that the triggers used in the reverse-engineered attacks using the proposed defense method are optimal, meaning they are the original triggers from our GhostEncoder technique. These triggers are employed to mitigate the impact of the backdoor in our generated backdoor encoder, resulting in an optimized clean encoder. We conducted two rounds of optimization on the model, and when the upstream dataset was CIFAR10, we defended the backdoor encoders of the downstream datasets STL10, GTSRB, and SVHN, obtaining clean triggers. Table 4 presents the test results of the clean triggers on the downstream datasets. It is evident that SSL-cleanse Zheng et al. (2023) effectively reduces the ASR of GhostEncoder. However, it significantly compromises the classification performance of the SSL encoder on normal inputs. Therefore, this defense method is not meaningful against GhostEncoder. Consequently, the SSL-Cleanse Zheng et al. (2023) method cannot directly address the global perturbation triggers of GhostEncoder attacks.

To corroborate the aforementioned analysis, we conducted tests on the backdoor encoders of BadEncoder and GhostEncoder before and after applying the SSL-Cleanse defense. For this purpose, we employed t-SNE visualization to examine the features of embedded triggers and clean inputs in the test set. Our experimental setup utilized the upstream pre-training dataset CIFAR10 and the downstream dataset STL10, as depicted in Fig. 8. In the case of BadEncoder (Fig. 8a and 8b), prior to the defense, the clusters of backdoor inputs (in black) and clean inputs from the target class (in red) exhibit a high degree of proximity in the feature space. However, after defense, a significant reduction in the aggregation of backdoor inputs is observed, as they move away from the target class. According to the outcomes presented in Table 4, GhostEncoder, after being subjected to the SSL-Cleanse defense, experiences a substantial decrease in its BA to 24.05%. This deterioration in classification performance results in a disorganized and unclustered representation as shown in Fig. 8d.

5.5.3. Other defenses

In addition to the aforementioned defenses, we also explored DECREE Feng et al. (2023), which can detect the presence of backdoors in models. The authors noted that DECREE failed to detect SSL-Backdoor and CTRL Li et al. (2023) attacks with $\mathcal{PL}^1$ -Norm around 0.23. The reason for the failure to detect SSL-Backdoor was its lower-than-expected ASR ($< 10\%$), which falls outside the expected ASR range ($> 99\%$). Compared to SSL-Backdoor and CorruptEncoder, our GhostEncoder employs more sample-specific and invisible triggers with higher $\mathcal{PL}^1$ -Norm. Additionally, GhostEncoder's ASR is not within the expected range of DECREE, making it undetectable by the DECREE defense method. While DECREE can detect the presence of backdoors, it cannot eliminate them. Therefore, we also explored the PatchSearch defense, which operates during the training phase and assumes poisoning-based backdoor attacks. However, this defense does not align with our threat model in

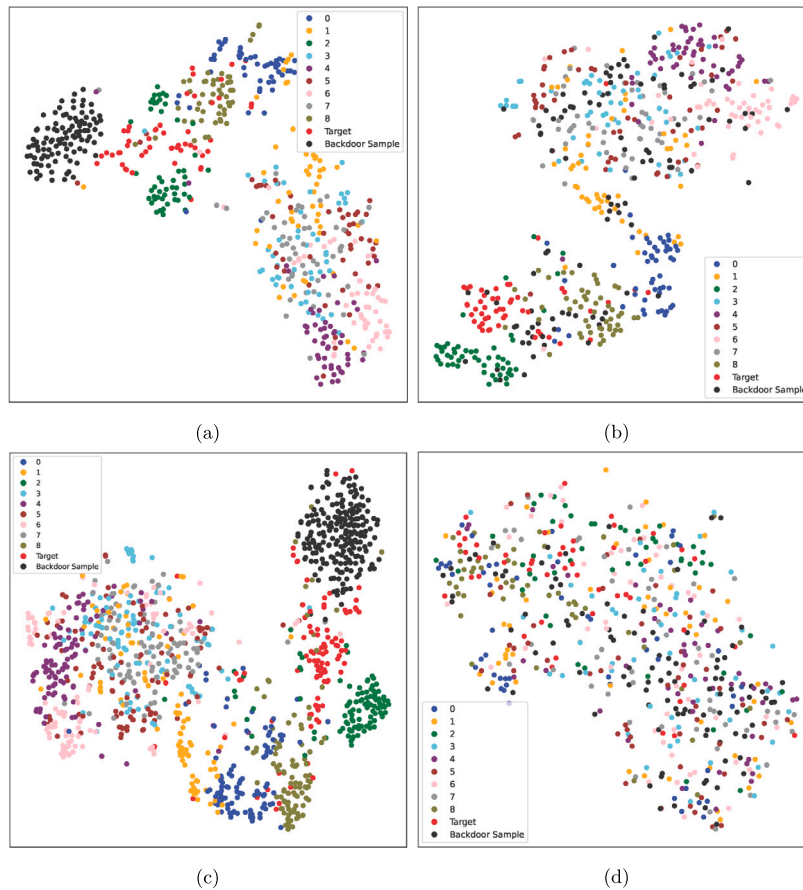


Fig. 8. t-SNE visualization of embedded triggers and clean input features in BadEncoder and GhostEncoder attacks (target class input: red; Embedded trigger input: black) (a) The results of the BadEncoder before the SSL-Cleanse defense. (b) The results of the BadEncoder after the SSL-Cleanse defense. (c) The results of the GhostEncoder before the SSL-Cleanse defense. (d) The results of the GhostEncoder after the SSL-Cleanse defense. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

this work since GhostEncoder utilizes weight modification-based attack methods instead.

6. Discussion

6.1. Why did the STRIP fail

The core concept of STRIP Gao et al. (2019) is to filter samples before feeding them into the model to detect the presence of a backdoor trigger. STRIP uses entropy as the basis for filtering, determining whether an image is suspicious by computing the entropy predicted by image classification. Specifically, STRIP overlays disruptions on the input image. If the input is a backdoor image, the classification outcome will be highly specific due to the trigger's presence, leading to very low entropy. Conversely, if the input is a clean image, the entropy will be relatively high. STRIP detects suspicious images by setting an entropy threshold, allowing it to differentiate between backdoor and clean images.

In supervised learning, due to the involvement of labeled information, models perform classification tasks by learning from these labels, enabling them to acquire more generalized features. The decision boundaries learned by these models tend to be clearer, ensuring that predictions for specific images do not deviate significantly from those for others. Consequently, the distribution of soft labels becomes more dispersed, and the entropy of clean images is relatively higher.

However, labeled data is absent in SSL, models must discern the relationship between triggers and samples to execute classification tasks. This can lead to fuzzier decision boundaries, less dispersed distributions of soft labels, and consequently, lower entropy for clean samples.

This results in indistinct threshold boundaries in the entropy distribution plots between backdoor and clean images. In such scenarios, the filtering mechanism of STRIP may mistakenly regard images as non-suspicious, leading to inaccurate detection of backdoor samples. Therefore, in the context of SSL, the defensive performance of STRIP significantly diminishes.

In contrast, our proposed detection framework, STRIP-Cl, focuses on the distribution of hard labels. Instead of using entropy distribution, we employ the maximum predicted distribution of hard labels for sample detection. Entropy has minimal influence on hard labels, allowing for a more precise reflection of the classifier's judgment concerning input images. Consequently, STRIP-Cl enhances the defensive capabilities of STRIP, broadens its defense scope, and enhances model security.

6.2. Insights from experimental observations

Under the parameter settings outlined in Section 5.1.2, we trained watermark encoders and observed that the effectiveness of generated watermarks is influenced to some extent by the stochastic nature of the training process. Through extensive experimentation, we elucidated the impact of watermark strength on the performance of attacks on downstream datasets, establishing a connection between this impact and the disparities in data distribution between pretraining and downstream datasets. Specifically, we found that watermark encoders with greater potency are better suited for generating watermark trigger patterns of a certain intensity when the data distribution of the downstream dataset is akin to that of the pretraining dataset. Conversely, in cases where there is a significant dissimilarity between the data distributions of upstream and downstream datasets, encoders of a less robust nature are

Table 5
ASR results of multiple attacks.

	SSAH-poison	SSAH-modify	ISSBA-poison	GhostEncoder
ASR	10.15	10.79	11.62	96.57

more adept at producing trigger perturbations. Building upon these insights, we conducted experiments wherein we tailored the training of respective watermark encoders suitable for backdoor attacks based on the distinct data distributions of various downstream datasets.

6.3. Other settings in self-supervised analysis

Existing backdoor attacks on SSL are only effective when using patch-based sample recognition triggers. To better understand backdoor attacks in SSL, we apply two existing attacks from supervised learning to our scenario: adversarial sample attack SSAH Luo et al. (2022) and poisoning-based ISSBA Li et al. (2021). Our adversarial sample attack has two settings: one follows the attack procedure of BadEncoder Jia et al. (2022) for the Image-on-Image attack used in this work, but replaces the backdoored samples generated by patch-based triggers with adversarial samples; the other is poisoning-based SSAH, where poisoned samples with added global triggers are mixed into the clean training set of the upstream encoder. Poisoning-based ISSBA aligns with the attack setting of poisoning-based SSAH. We then evaluate the ASR of downstream classifiers trained from the backdoored encoder. The results are shown in Table 5.

From the experimental results, it can be observed that adversarial sample attacks and targeted poisoning attacks on pre-trained self-supervised encoders are difficult to succeed. These attacks can be successful and covert in supervised learning because there is a specific target label that provides strong hints during the attack process. However, SSL only considers positive pairs or negative pairs and lacks obvious features (such as patch-based triggers). Therefore, it is challenging for such sample triggers to establish strong correlations between victim images and target images.

6.4. Limitations of current defenses

Upon assessing the efficacy of current backdoor defense mechanisms against GhostEncoder, it became apparent that their limitations are primarily due to a focus on specific backdoor attack types. GhostEncoder, however, introduces a more intricate and concealed attack strategy. The central strategy of STRIP Gao et al. (2019) and STRIP-CL is the detection and neutralization of patch disturbances triggered by a singular, invariant stimulus. These triggers persistently affect classification when merged with random, uncontaminated images, leading to distinct classification biases. GhostEncoder employs a comprehensive perturbation technique, impacting every pixel rather than confined regions. This comprehensive trigger significantly alters the classification when merged with uncontaminated images, rendering these defenses ineffective in distinguishing GhostEncoder's extensive perturbations.

SSL-Cleanse Zheng et al. (2023), while effective against certain SSL encoder-targeted backdoor attacks like SSL-Backdoor Saha et al. (2022) and ESTAS Xue and Lou (2022), proves inadequate against GhostEncoder. The global perturbations of GhostEncoder extend across the full image space, affecting every pixel. This complexity, compounded by the trigger's sample-specific and covert nature, significantly hampers the reverse identification of a unified trigger. Evidence from Table 4 indicates that SSL-Cleanse fails in later defense stages, even when employing actual triggers from the pre-trained dataset as the reverse-engineered unified trigger.

Furthermore, the DECREE Feng et al. (2023) defense, designed to identify backdoors within models, was unsuccessful in detecting backdoor attacks with a low $\mathcal{PL}^1$ -Norm such as SSL-Backdoor and CTRL Li et al. (2023). GhostEncoder utilizes a trigger that is not only more

concealed and sample-specific but also has a higher $\mathcal{PL}^1$ -Norm, which lies outside DECREE's anticipated detection threshold, highlighting its limitations in identifying novel and intricate backdoor attacks. Lastly, PatchSearch Tejankar et al. (2023), conceptualized for poison-type backdoor attacks, is ineffective against GhostEncoder, which manipulates model weights, diverging from PatchSearch's assumed defense scenario.

To conclude, while current defenses effectively mitigate some backdoor attacks, they are less efficacious against innovative forms like GhostEncoder, characterized by global reach, concealed triggers, and sample-specific traits. This underscores the need for more inclusive and adaptable strategies in backdoor defense, geared towards addressing the increasingly varied and complex landscape of backdoor attacks.

7. Conclusion

In this work, we propose the first dynamic covert backdoor attack against SSL, known as GhostEncoder. We demonstrate that existing backdoor attacks are susceptible to mitigation by current defense mechanisms primarily due to the visibility or static nature of their trigger mechanisms. Building on this understanding, we explore a novel trigger pattern and apply it for the first time to non-poisoned backdoor attacks. Experimental results confirm that our GhostEncoder deceives victim models with a high attack success rate while maintaining model utility. Furthermore, it is evident that the existing defenses against backdoor attacks are insufficient to counter our approach. We investigate the reasons behind STRIP's failure in the self-supervised setting and propose a novel enhancement, termed STRIP-CL. Interesting avenues for future research encompass: 1) Extending our attack to other domains within self-supervised scenarios, such as natural language processing; 2) Devising novel defense strategies to thwart our attack.

CRedit authorship contribution statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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